

SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

Dept: CSE - AIDS | Course: DSA06 - Data Handling and Visualization

Set 2: Customer Feedback Analysis

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Dataset: Customer Satisfaction — Customer ID, Age, Satisfaction Score for 5 customers.

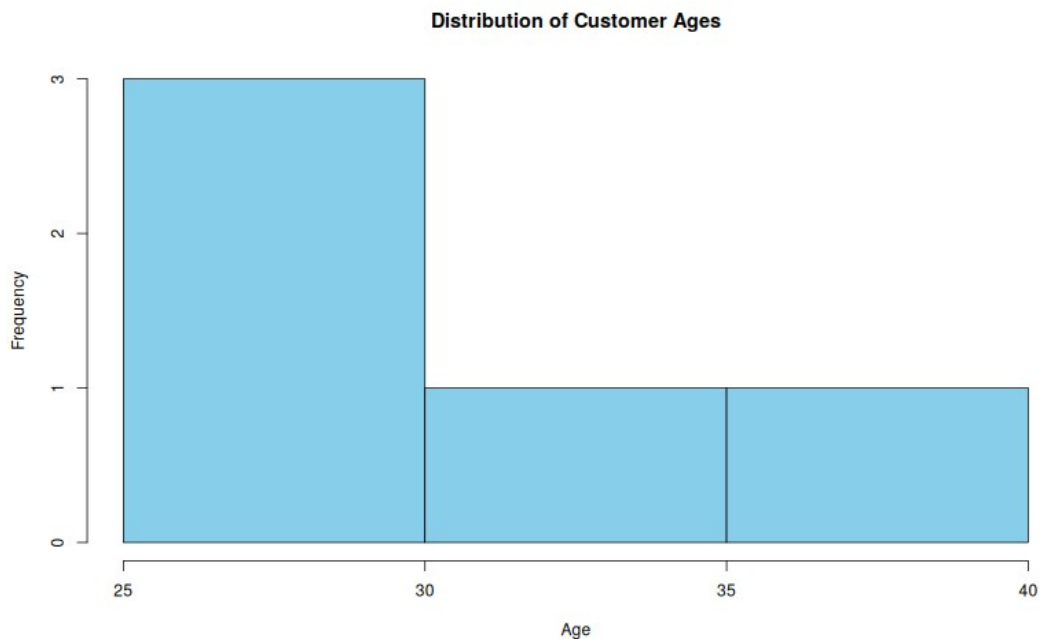
Question 1

Create a histogram to represent the distribution of customer ages. Label the axes and title the chart.

Code

```
cust <- data.frame(  
  CustomerID = 1:5,  
  Age = c(25,30,35,28,40),  
  Satisfaction = c(4,5,3,4,5)  
)  
  
hist(cust$Age, col="skyblue", breaks=4,  
     xlab="Age", ylab="Frequency",  
     main="Distribution of Customer Ages")
```

Output



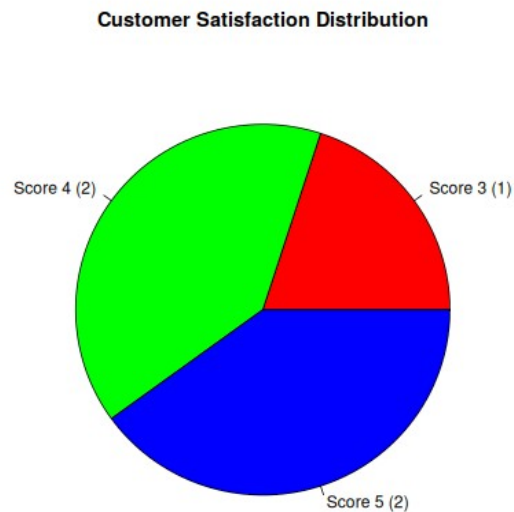
Question 2

Generate a pie chart to display the overall distribution of customer satisfaction scores. Include labels.

Code

```
tab <- table(cust$Satisfaction)
pie(tab, labels=paste0("Score ", names(tab), " (", tab, ")"),
    col=rainbow(length(tab)), main="Customer Satisfaction Distribution")
```

Output



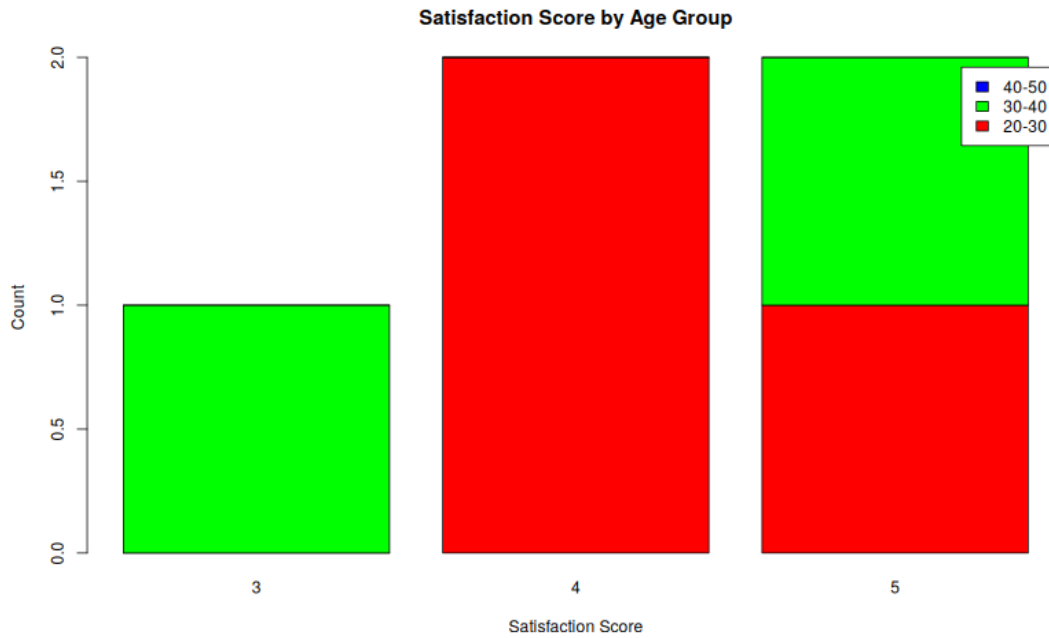
Question 3

Build a stacked bar chart to visualize the distribution of customer satisfaction scores by age group.

Code

```
cust$AgeGroup <- cut(cust$Age, breaks=c(20,30,40,50),
    labels=c("20-30", "30-40", "40-50"))
tab2 <- table(cust$AgeGroup, cust$Satisfaction)
barplot(tab2, col=rainbow(nrow(tab2)), legend.text=TRUE,
    xlab="Satisfaction Score", ylab="Count",
    main="Satisfaction Score by Age Group")
```

Output



Question 4

Develop a word cloud from open-ended customer feedback to identify prevalent customer sentiments.

Code

```
feedback <- c("Great service and fast delivery",
              "Product quality was excellent",
              "Support staff were very helpful",
              "Delivery was late but product is good",
              "Excellent quality and great support")
library(tm); library(wordcloud)
corpus <- Corpus(VectorSource(feedback))
corpus <- tm_map(corpus, content_transformer(tolower))
corpus <- tm_map(corpus, removePunctuation)
corpus <- tm_map(corpus, removeWords, stopwords("english"))
tdm <- as.matrix(TermDocumentMatrix(corpus))
freq <- sort(rowSums(tdm), decreasing=TRUE)
wordcloud(names(freq), freq, min.freq=1, colors=brewer.pal(8,"Dark2"))
```

Output



Insight: words like 'excellent', 'quality', 'support', 'great', and 'delivery' appear largest since they repeat most often in the feedback, showing that customers' sentiment leans strongly positive around product quality and support.